ANALYSIS PLAN FOR COHORT IPT-TRIAL SHORT-TERM FOLLOW-UP

Graduating To Resilience in Kamwenge, Uganda, Cohort 2, IPT-Trial
Pre-Analysis Plan

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Implementing Partner: AVSI Foundation

Introduction

This document outlines the plan for analysis of a short-term follow-up survey that is part of the project "Graduating To Resilience in Kamwenge, Uganda." The survey follows up with a sample of study participants of the "IPT Trial" for which Cohort 2 program participants were randomly assigned to additional Group Interpersonal Therapy (G-IPT) programming (= "Treatment").

Empirical analysis

The analysis will be based on OLS regressions of outcomes of interest on a treatment indicator, controls for stratification controls and baseline values of outcomes where available or their closest proxies. The analysis will be carried out separately for host and refugee communities and with the pooled sample of respondents from both communities. In addition we'll analyze heterogeneity of treatments effects using the ML techniques described in Chernozhukov et al (2020)[1] with a set of baseline variables that will include baseline mental health (Kessler 6, number of mentally unhealthy days), household wealth (measured as the sum of value of land owned, livestock and durable assets) and others. Outcomes will be winsorized at the top and bottom 1% if the standard deviation of the unwinsorized outcome exceed the winsorized version of the outcome by more than 50%.

Indices

For the construction of outcome group summary measures based on components with different scales, we will use the methodology detailed in by Kling, Liebman, and Katz (2007), unless the index is a specific index in the academic literature, in which case we will use the method employed in that literature to compute the index. For those concepts without a preconceived index formula, our methodology consists in first signing all variables consistently such that higher is telling a consistent story for the index. Then, we standardize the individual components

[1] Chernozhukov, Demirer, Duflo, Fernandez-Val (2020): "Generic machine learning inference on heterogenous treatment effects in randomized experiments," ArXiv: 1712.04802.

of the index, by subtracting the comparison group mean and dividing by its standard deviation. Then, we take the average of the now-standardized components into a single measure, and then again finally standardize the average (again to the comparison group mean and standard deviation). We will use the follow-up survey control group values for the standardization.

Primary outcomes of interest

We have two sets of primary outcomes – psychological and economic ones.

1. **Psychological distress**
    - Summary measure: z-score index of three components
    - Kessler-10 score (10-50)
    - PHQ-9 score (0-27)
    - GAD-7 score (0-21)

2. **Health-Related Quality of Life (BRFSS)**
    - Summary measure: z-score index of three components
    - Healthy Days Index := 30 - (sum of 2 items: physically unhealthy and mentally unhealthy days) (0 - 30)
    - Subjective health and mental health (average of 2 items)
    - Days of good functioning (0-30) (1 item)

3. **Economic activity index**
    - Summary measure: z-score index of three components
    - Primary participant labor supply index (z-score)
    - Investments (5th root of total)
    - Expansion intentions index (0 - 4)

Additional details on construction of outcomes in the economic activity index:

<u>Labor supply</u>

- *Summary measure*: Z-score index constructed from two subcomponents
- Components: 1. sum of number of days worked by the primary target participant in farming, livestock, business in the past 30 days; 2. sum of hours worked in farming, livestock, business in the past 7 days

<u>Investments</u>

- *Summary measure*: $y^{1/5}$ where y = sum of farming, livestock and business investments in past four months[2]

[2] The power transformation is applied due to the extreme right skew of the investment outcome.

- Components: Totals for each of three sets of questions that asked for how much money, from different sources, respondents had invested in farming activities, livestock and business activities over the past four months

Economic expansion intentions:

- Summary measure: Sum of five binary indicators
- Components: 1. plans to invest more money into developing farming business; 2. plans to purchase land in the next 12 months; 3. plans to invest more money into developing livestock business; 4. plans to start a new business in the next 12 months OR (if has existing business) plans to invest more money into developing off-farm businesses in the next 12 months

Secondary outcomes

All outcomes that are not listed as primary outcomes are considered secondary outcomes. These variables will be derived from the attached survey instrument and are not listed individually in this document.

Data collection and timing of analysis

While data collection was completed at the time of submission of this analysis plan, the researchers have been blind to the treatment effect estimates, with the exception of analysis of survey attrition which was conducted during data collection and included comparisons of attrition rates by treatment status for the purpose of management of potential differential attrition.